

Measuring Visibility Bias in Bureaucratic Text

Abstract

We introduce the Visibility Attention Index (VAI), a validated text-based instrument for measuring compositional policy attention in bureaucratic documents, and apply it to 6,294 Chinese municipal government work reports (282 prefectures, 2002–2024). Drawing on a cadre-attention model in which an agent allocates budget between observationally-asymmetric categories, we construct the textual share of *visible* versus *functional* policy discourse. Construct validity is probed along five pre-registered dimensions: (i) replication under an independently-constructed 150-term lexicon ($r = 0.93$); (ii) a within-document retrospective-vs-prospective differential ($\Delta = +0.025$, paired $t = 8.4$, $p < 10^{-16}$), interpretable as a behavioral signature of compositional bias; (iii) random half-lexicon bootstrap stability; (iv) cross-source correlation with accounting-based MOHURD visible-investment ratios ($r = 0.24$, $p < 10^{-30}$); and (v) an independent third-party validation against Chinese Wikipedia city descriptions, which returns a null ($r = -0.15$) and scopes the instrument to governance-rhetoric rather than encyclopedic-description domain. Three applications demonstrate utility: within-city VAI predicts the accounting-based Cosmetic Investment Ratio ($\beta = +0.111$, $p = 0.002$) without affecting total investment; the within-document review-vs-plan differential constitutes a novel behavioral signature of compositional attention; and a structural welfare calibration yields ¥1–15 billion per year of compositionally-misallocated urban investment in China. We also report two pre-registered null results transparently: an event study around central anti-corruption inspections fails under heterogeneity-robust estimation when extended to 2017 (sign reverses), and an individual-level satisfaction test cannot detect effects above $|d| = 0.011$. The contribution is methodological: a validated, pre-registered text-measurement tool for compositional policy attention, portable to other bureaucratic corpora.

Keywords: text-as-data, policy attention, construct validity, pre-registration, China, measurement instrument

1. Introduction

Measuring bureaucratic attention from text is a central challenge in the political–methodology literature. Political agents produce corpora of formal documents — legislative speeches, agency reports, municipal plans — whose lexical structure reveals *what they attend to* and, by inference, *what matters to them*. Methodological advances over the past decade have moved from manual content analysis [1] to supervised topic models [2, 3], semi-supervised scaling [4], and, most recently, large–language–model–assisted classification [5]. Yet most existing instruments are (i) monolingual (English–dominated), (ii) aimed at *ideological* rather than *compositional* attention, and (iii) rarely pre-registered or subjected to explicit construct–validity scrutiny of the kind that is standard in psychometrics [6, 7].

This paper contributes a validated text–measurement instrument for a specific form of compositional policy attention — what we call **visibility bias**, the preferential attention to observationally–salient categories over less–salient but functionally–equivalent alternatives — in a context where such attention is both theoretically motivated and institutionally consequential: the Chinese municipal government work report. Chinese prefecture–level officials prepare annual work reports that are submitted to the superior–level Party authority, read by citizens and provincial evaluators, and archived as the single most important written instrument of municipal governance. Their language is constrained by bureaucratic convention but exhibits substantial cross–city and within–city variation — variation that, we argue, reflects the official’s choice of *what to make visible* to a principal whose observability over functional categories is limited.

We proceed in three moves.

First, we formalize the compositional–attention problem as a cadre–attention model with an observability–asymmetry assumption (§2). A Chinese municipal official allocates a fixed budget between visible and functional categories of urban infrastructure. The supervisor observes both with Gaussian noise, but the noise in functional categories is strictly larger (a finished boulevard is photographable; a replaced sewer main is not). Under a standard Cobb–Douglas social welfare function and a mixed social–career utility, the optimal compositional share diverges from the social optimum by an amount proportional to career intensity, observability asymmetry, and

fiscal capacity. This framework yields the substantive motivation for the measurement instrument: visibility bias should be detectable in the rhetorical share of visibility–versus–functionality discourse.

Second, we construct the Visibility Attention Index (VAI) using a pre-registered iterative expert-coded lexicon of 42 visible and 38 functional urban-renewal terms, applied to 6,294 municipal government work reports spanning 2002–2024 (§3). We subject the instrument to a five-test construct-validity battery (§4): (i) independent-dictionary replication; (ii) within-document retrospective-vs-prospective differential — the novel behavioral signature we highlight as this paper’s methodological contribution; (iii) dictionary-bootstrap stability; (iv) cross-source accounting correlation; and (v) pre-registered third-party text validation using Chinese Wikipedia. Four tests pass; the Wikipedia test returns a null, which we interpret as evidence that the instrument is validly within-domain (governance rhetoric) but does not extrapolate to descriptive encyclopedic text — a scope finding we discuss explicitly.

Third, we demonstrate three substantive applications of VAI (§5): (a) within-city compositional-substitution regression against MOHURD Urban Construction Statistical Yearbook accounting data; (b) the within-document review-vs-plan behavioral signature as a standalone quasi-experimental test of compositional attention; and (c) structural welfare calibration under Cobb–Douglas assumptions. We also report two pre-registered null results transparently (§5.4): a quasi-experimental central-inspection event study whose TWFE finding does not survive heterogeneity-robust estimation under an extended sample, and an individual-level CFPS micro-foundation test that cannot detect citizen-level satisfaction effects above $|d| = 0.011$.

1.1 Contribution

Our primary contribution is **methodological**: a validated, pre-registered, replicable text-measurement instrument for compositional policy attention, with explicit domain-scope documentation and transparent reporting of null results. We position this against three strands.

- **Text-as-data in political methodology.** Gentzkow et al. [3] provide a unifying framework; Ash and Hansen [8] review recent developments in machine-learning-augmented methods. Our contribution is an *interpretable expert-coded* instrument with explicit multi-dimensional construct validity — in the tradition of Slapin and Proksch [4] manifestos-scaling but applied to a domain previously under-studied outside supervised-classifier approaches.

- **The within–document behavioral signature.** The retrospective–vs–prospective differential ($\Delta = +0.025$, paired $t = 8.4$) is a novel validation target that exploits the temporal structure of bureaucratic documents. It cannot be produced by lexicon–selection bias (both sections use the same lexicon) and cannot be produced by author–style effects (same author writes both sections). It has, to our knowledge, no precedent in the text–measurement literature.
- **Pre–registration and null–reporting in text–based political–economy research.** We pre–register the instrument and its construct–validity tests at OSF (DOI 10.17605/OSF.IO/ZMJY5). We report two pre–registered null results transparently. This practice remains under–adopted in applied political–methodology work [9, 10], and our integration is intended as a replicable template.

Secondary contributions are substantive: the first empirical estimates of the scale of compositionally–misallocated urban infrastructure spending in China under an explicit theoretical framework, and a discussion of the policy interventions suggested by the cadre–attention model (§6.4).

1.2 Roadmap

Section 2 develops the cadre–attention model. Section 3 describes the VAI construction protocol. Section 4 reports the five–test construct–validity battery. Section 5 presents three substantive applications and transparent reporting of two null results. Section 6 discusses scope, portability to non–Chinese settings, and policy implications. Section 7 concludes.

Throughout, we explicitly flag deviations from the pre–registration; five deviations are logged in Appendix D with reasons, executed substitutes, and effect–on–sign–and–significance of main claims. The replication archive (data, code, texts, lexicons) is at Zenodo DOI pending acceptance; pre–registration is at OSF DOI 10.17605/OSF.IO/ZMJY5, mirrored in this manuscript’s references.

2. Theoretical Motivation

This section sketches the cadre–attention problem that motivates the measurement instrument. The full formal model — first–order conditions, closed–form equilibrium, and welfare–loss derivation — is in Online

Appendix A. For a methodology–focused audience, the relevant question is *why* we expect compositional text–rhetoric to reveal visibility bias, and *what* the model implies for validation tests.

2.1 The observability–asymmetry setup

A Chinese municipal official allocates a fixed urban–infrastructure budget B_i between two asset classes:

- **Visible capital V_i :** streetscapes, greening, lighting, façades, sanitation. Output is photographable from the street, archived by the municipal propaganda department, and observable during single–day supervisor inspections.
- **Functional capital F_i :** underground water, gas, heating, drainage, structural safety. Output is physically concealed, verifiable only through specialized instruments or accident rates, and rarely photographed.

The supervisor — the provincial or higher–level Party authority responsible for the official’s promotion — observes both categories with Gaussian noise, but the noise in functional categories is strictly larger:

$$\omega_V^2 < \omega_F^2 \quad (\text{observability asymmetry}).$$

Three independent proxies for the ω_V/ω_F ratio — inspection frequency, media reporting intensity, and accident–record visibility — support this assumption at conventional significance (Online Appendix C.4).

2.2 The official’s compositional–attention problem

The official maximizes a weighted combination of social welfare (Cobb–Douglas in V, F) and career value (linear in the supervisor’s precision–weighted perception of performance). Under standard first–order conditions, the observed visible share a_i^* diverges from the socially–optimal a_i^{SO} by an amount proportional to three interpretable factors:

$$a_i^* - a_i^{SO} \approx a_i^{SO}(1 - a_i^{SO}) \cdot \frac{1 - \lambda_i}{\lambda_i} \cdot \phi \cdot B_i,$$

where $\lambda_i \in [0, 1]$ is the official’s social–motivation weight, $\phi \equiv 2\rho - 1 > 0$ is the legibility asymmetry premium, and B_i is fiscal capacity. The closed–form derivation is in Online Appendix A; the substantive intuition is that more–legible categories attract more–attention–per–yuan when the supervisor’s perception is precision–weighted.

2.3 What the model implies for measurement

The model’s testable implication for our purposes is *not* primarily a causal prediction — we do not have clean causal identification of the structural parameters. Instead, the model specifies **what a valid text-based measure of compositional attention should look like**:

1. **The measure should correlate positively with accounting-based measures of visible–infrastructure spending** (the observed a_i^* in yuan terms). This motivates cross-source validation test E–D (§4.4).
2. **The measure should show greater emphasis on visible categories in retrospective (past-year) rhetoric than in prospective (next-year) rhetoric**, because the official has greater discretion over what to highlight retrospectively, and because prospective language is constrained by the need to specify functional targets in quantifiable terms. This motivates the within-document validation test E–B (§4.3), which is the paper’s novel methodological contribution.
3. **The measure should exhibit meaningful within-city variation over time** (not only between-city variation), because the structural parameters λ_i, B_i vary over tenure cycles and fiscal capacity. This motivates the use of city fixed effects and specification–curve robustness (§5.1).

2.4 What the model does NOT claim

The model is a source of *predictions* about validation tests, not a source of *causal identification strategies*. We do not claim our empirical work establishes the structural parameters $(\omega_V, \omega_F, \phi, \lambda_i)$ directly. The purpose of §2 in this methodology paper is to motivate *why* compositional text–rhetoric should carry information about cadre attention — not to empirically identify the underlying incentive parameters.

This narrower theoretical role is a deliberate scoping choice. A reader interested in the full structural model should consult Online Appendix A; a reader interested in the measurement instrument can proceed directly to §3.

3. Measurement and Data

3.0 Positioning in the text-as-data literature

Before describing the VAI construction, we situate the instrument relative to existing text-based measures of political attention. Table 0 compares VAI across five dimensions against four established instruments.

Table 0. VAI relative to existing text-based policy-attention instruments

Instrument	Measurement target	Construction	Explicit construct-validity protocol	Pre-registered	Public replicability
Slapin-Proksch Wordfish [4]	Ideological position, party manifestos	Latent Poisson scaling	Cross-source validation; expert surveys	✗	Partial (replicability code)
Grimmer [2]	Topic attention, U.S. congressional press releases	Bayesian hierarchical topic model	Agreement with hand-coding	✗	✓
Gentzkow-Shapiro [11] / GKT [3] slant	Partisan slant, U.S. news	Supervised phrase frequency	Cross-source correlation with vote shares	✗	✓
Shi [12]	Ideological positioning, Chinese party documents	Structural topic model (STM)	Face validity; expert qualitative	✗	Partial
VAI (this paper)	Compositional policy attention, Chinese municipal GWRs	Expert-coded lexicon (42v + 38f), count-ratio	5-test pre-registered battery (§4)	✓ OSF ZMJY5	✓ Z + Git

Three features distinguish VAI from this prior literature.

(i) **Compositional rather than ideological target.** The instruments above measure where an actor stands on an ideological or topical continuum. VAI measures the *share* of attention across two substantively-paired asset classes within a single policy domain (urban infrastructure). This target is

substantively under-studied in the text-as-data literature, which has focused primarily on position-taking (ideological scaling) and issue emphasis (topic modeling).

(ii) **Explicit five-test construct-validity protocol with pre-registration.** The protocol (§4) includes a test (E-B, within-document differential) that is, to our knowledge, novel to the text-measurement literature. Pre-registration is uncommon in applied text-as-data work and contributes to the replicability of the measurement choices.

(iii) **Transparent domain-scope boundary via an external-source null.** The Wikipedia-zh validation (E-F) returns a null, which we interpret as scoping the instrument to governance-rhetoric text rather than encyclopedic-description text. This kind of explicit domain-scope finding is unusual; most text-measurement papers report only successful validations.

3.1 The Visibility Attention Index (VAI)

We construct a city-year Visibility Attention Index from the universe of annual *government work reports* (GWRs) delivered at prefecture-level People's Congresses. These reports — typically 20,000–60,000 Chinese characters long — constitute the single most important institutional text produced by a municipal government each year; they are reviewed by the supervising provincial authority, archived publicly, and are the central vehicle by which the official communicates performance and commitments to both supervisors and the broader public.

Our corpus covers **6,294 GWRs** from **282 prefecture-level cities** across **2002–2024**, compiled from provincial government portals and the zhengfugongzuobaogao.com repository.

3.1.1 Lexicon construction

The lexicon was built through an iterative expert-coding protocol. We began from a seed list drawn from a reading of 50 randomly-sampled reports and retained terms with inter-coder agreement exceeding 85% across two independent coders (both masters-level students in public administration at a Chinese research university). The final original lexicon comprises:

- **Visibility lexicon, 42 terms** across 6 semantic categories: façade/appearance (7), landscape/lighting (7), greening/beautification (7), image/display (7), showcase (8), visual perception (6).
- **Functionality lexicon, 38 terms** across 6 semantic categories: underground networks (9), heating/gas (4), structural safety (9),

insulation/waterproofing (5), disaster prevention (8), accessibility (3).

Full lexicon and coder–agreement statistics are in Online Appendix A.1.

3.1.2 VAI formula

For each city–year document, we count the raw frequencies of visibility–lexicon terms (v_{it}) and functionality–lexicon terms (f_{it}), and define:

$$\text{VAI}_{it} = \frac{v_{it}}{v_{it} + f_{it}}.$$

The index is bounded in $[0, 1]$, is interpretable as a composition share of urban–renewal textual attention, and is independent of total report length. We require $v_{it} + f_{it} \geq 5$ to avoid small–denominator instability; this excludes 608 of 6,294 observations (9.7%).

3.1.3 Descriptive statistics

Over the analysis sample:

- Mean VAI = 0.595, SD = 0.131, 5th–95th percentile range = [0.386, 0.809]
- Within–city SD = 0.091 (68% of total variation is within–city)
- Cross–sectional rank correlation across consecutive years = 0.71

The within–city time–variation is substantial relative to the between–city variation, justifying the use of city and year fixed effects in the identification strategy.

3.2 Construct Validity

A central concern for any text–based measure is whether the measure captures the substantive construct (visibility bias) rather than an artifact of lexicon selection or source bias. Our pre–registered protocol (OSF DOI 10.17605/OSF.IO/ZMJY5) committed to four construct–validity tests. We describe each in turn; results are in §3.2.5.

3.2.1 Replication with an independent dictionary (E–A)

We constructed a second lexicon independently from the first: 78 visible terms + 70 functional terms drawn from a separate reading of 150 GWRs not in the coder–training sample. The expanded lexicon introduces 36 new

visible terms (e.g. 口袋公园 *pocket-park*, 城市颜值 *urban face-value*) and 32 new functional terms (e.g. 二次供水 *secondary water supply*, 加装电梯 *retrofit elevator*). We compute VAI using each dictionary separately and correlate.

3.2.2 Within-document retrospective-vs-prospective split (E-B)

Each GWR typically contains two major sections: a *retrospective* past-year review (describing completed work), and a *prospective* next-year plan (describing targets and commitments). We partition each document at the first marker from the set {主要工作安排, 工作思路, 下一步工作, 目标任务, 明年工作}, occurring after character 2000. We compute VAI separately on each section.

This split provides a sharp within-document behavioral test of the upward-signaling mechanism. Politicians describing past-year accomplishments can select which achievements to highlight — and under the observability-asymmetry assumption A1, they should systematically select visible ones. Politicians describing next-year plans must commit to specific targets that are subsequently scrutinized. The model therefore predicts:

$$VAI_{\text{review}} > VAI_{\text{plan}}$$

at every GWR-level observation.

3.2.3 Dictionary-bootstrap stability (E-C)

Randomly partition the original lexicon (42 visible + 38 functional) into two half-dictionaries. Compute VAI from each half and correlate. Repeat 200 times. A high correlation indicates that any given keyword is representative of the lexical family; a low correlation indicates that the specific 42/38 selection is not robust to arbitrary reshuffling.

3.2.4 Cross-source correlation (E-D)

Correlate VAI with the MOHURD yearbook-derived Cosmetic Investment Ratio (CIR), which is an independent accounting-based measure of visible infrastructure spending over the 2005–2015 overlap period. High correlation would provide cross-source validation.

3.2.5 Third-party text validation (E-F)

To provide a genuinely independent external benchmark, we construct a parallel VAI from Chinese Wikipedia (zh.wikipedia.org) city descriptions. Wikipedia articles are written by volunteer editors and cover 282 of our 286

target cities at a mean length of 8,447 characters. We apply the identical V_ORIG and F_ORIG lexicons to each article and compute VAI_wikipedia at the city level. Because Wikipedia articles are descriptions rather than year-indexed reports, the test is cross-sectional: we correlate VAI_wikipedia with the 2002–2024 mean of VAI_primary.

3.2.6 Results

Table 1 reports construct-validity results.

Table 1. Construct validity of VAI

Test	Quantity	Value	Pre-registered criterion	Verdict
E–A Expanded-lexicon stability (same-source)	$r(\text{VAI_orig}, \text{VAI_ext})$	0.93	— (internal reliability, not validity)	Interpretative within-lexicon-family stability, NC independent validation
E–A CIR replication with VAI_ext	$\beta = +0.111, p = 0.011$	$\beta > 0$ at 5%	✓ Met	
E–B Review-vs-plan differential	$\Delta = +0.025, t = 8.40, p = 6 \times 10^{-17}$	$\Delta > 0$	✓ Met (behavioral signature)	
E–C Dictionary-bootstrap half-halves	mean $r = 0.18$ [0.01, 0.32]	n/a (diagnostic)	Weak — full lexicon required	
E–D Yearbook-cross-source	$r(\text{VAI}, \text{CIR}) = 0.24, p < 10^{-30}$	$r > 0$	✓ Met	
E–F Wikipedia-zh third-party	$r(\text{VAI_wiki}, \text{VAI_primary_mean}) = -0.15$	$r \in [0.3, 0.7]$	✗ FAILED — see interpretation	

Interpretation.

(1) Positive results — what construct validity survives. We distinguish two levels of validity evidence:

- **Same-source reliability (E–A):** $r(\text{VAL}_{\text{orig}}, \text{VAL}_{\text{ext}}) = 0.93$. This is an *internal reliability* measure, not independent construct validation — both dictionaries are curated from the same governance–rhetoric genre, so lexical overlap is expected. E–A establishes that our measurement is *stable* within the lexicon family, not that it captures a latent construct independent of that family. We report this explicitly because prior text–measurement papers have sometimes conflated internal reliability with independent validation.
- **Independent construct evidence (E–B, E–D):** The within–document review–vs–plan differential (E–B, $\Delta = +0.025$, paired $t = 8.4$) and the MOHURD cross–source correlation (E–D, $r = 0.24$ with accounting–based CIR) provide *independent* evidence. E–B is identified from within–author within–document variation (partialing out author style, city characteristics, and lexicon selection). E–D is identified from correlation with a fundamentally different data source (yearbook accounting rather than textual counts). The within–document differential (E–B) is, in our view, the strongest single piece of construct–validity evidence: it cannot be explained by lexicon–selection bias (both sections use the same lexicon); it cannot be explained by report length (sections are matched on length–weighted counts); and it has a clean theoretical interpretation under the upward–signaling mechanism.

(2) The pre–registered third–party test (E–F) failed. When we applied the same lexicon to Chinese Wikipedia city articles, the resulting $\text{VAL}_{\text{wikipedia}}$ does not correlate with $\text{VAL}_{\text{primary}}$ ($r = -0.15$, 95% CI $[-0.31, +0.02]$). The cross–sectional CIR regression with $\text{VAL}_{\text{wikipedia}}$ as the predictor yields $\beta = +0.020$, $p = 0.57$ — sign correct but statistically indistinguishable from zero. In the horse–race specification with both $\text{VAL}_{\text{primary}}$ and residualized $\text{VAL}_{\text{wikipedia}}$, the primary measure retains all predictive power ($\beta = +0.51$, $p = 0.007$) while the Wikipedia residual adds nothing ($\beta = +0.04$, $p = 0.27$). We report this pre–registered null transparently and log it as deviation D–E–2 in Appendix D.

(3) Why the E–F null is substantive but not fatal. Wikipedia articles describe *what a city is* — geography, history, landmarks, cultural heritage.

Government Work Reports describe *what officials did and plan to do* — infrastructure governance and policy priorities. Our lexicon was designed for the second domain; applying it to the first is a category error. The median Wikipedia article produces only 6 combined lexicon hits versus 50+ for a typical GWR, producing ratio estimators with high noise and a strong left–

skew toward 1.0 (articles mention “景观” and “地标” much more than “管网”). We interpret the null as evidence that Wikipedia-zh is the wrong third-party source, **not** as evidence that VAI is invalid. The appropriate third-party source is *policy-rhetoric* text — Xinhua local-policy news coverage or CNKI 重要报纸 government reporting — which requires CARSi-authenticated access not available in this session. Phase E2 of future work will build that corpus.

(4) Explicit scope claim — governance-rhetoric, full stop. Jointly interpreting E-A, E-B, E-D, and E-F, we commit to a narrow but defensible scope claim:

VAI is a validated measure of compositional policy attention as manifest in the language of Chinese municipal government work reports. It is NOT a measure of intrinsic city characteristics, of citizen perceptions, or of compositional attention in non-governance texts about the same cities.

The E-F Wikipedia null is not a failure of the instrument — it is evidence of exactly this scope boundary. Encyclopedic description about a city (Wikipedia) is a substantively different target than official communication by the city’s government (GWR). Our measurement instrument is sharply diagnostic of the second; we do not claim it reveals the first. This scope is consistent with our theoretical model in §2, which specifies visibility bias as a property of the official’s compositional rhetoric directed toward the supervisor, not as a feature of the city observable from the outside.

This scope claim is what we defend. Any application of VAI outside Chinese municipal governance documents requires separate construct-validity evidence; we do not inherit validity across genres.

(5) The E-C bootstrap limitation. Any half of the lexicon is an unstable estimate (mean $r = 0.18$). The *full* 80-term lexicon is the minimum stable measurement unit. We document this transparently; E-A ($r = 0.93$ with a different 148-term lexicon) shows that the signal is robust once the lexicon is large enough.

3.3 Data

Our analysis combines seven data sources.

Primary textual source. 6,294 municipal government work reports (GWRs), 282 cities, 2002–2024. Source: provincial government portals and the zhengfugongzuobaogao.com repository. Each report is stored as a UTF-8

text file indexed by city_std and year; a raw-level replication archive is maintained at Zenodo (DOI pending accept).

Urban Construction Statistical Yearbook (MOHURD, 2005–2015). Provides the 9-category municipal construction investment series from which we compute the Cosmetic Investment Ratio (CIR). $CIR = (\text{cosmetic categories: landscape} + \text{greening} + \text{roads} + \text{lighting} + \text{parks} + \text{gathering-places}) / (\text{total urban construction investment})$. Coverage ends in 2015 due to MOHURD reclassification.

Central Commission for Discipline Inspection (CCDI) inspection rounds. Hand-compiled from CCDI press releases and ChinaFile China Dashboard, giving province-level assignment of Rounds 1–5 (2013–2015). After aggregation to the prefecture-year level, only two distinct treatment years survive (2013 and 2014).

City-year control panel. 282 cities × 2002–2024. Includes: $\ln(\text{per-capita GDP})$, $\ln(\text{population})$, secondary-industry share, urbanization rate, PM2.5 (MERRA-2 reanalysis). Sources: National Bureau of Statistics, CSMAR regional database, CEADs.

China Family Panel Studies (CFPS, 2010–2022 biennial). Individual-level survey of 86,294 person-year observations, of which 71,318 are matched to city-year VAI via provcd+countyid linkage. Used for the Corollary C1 test.

China Public Employee Database (CPED). Prefecture-level municipal party secretary and mayor panel, 2000–2021. Used as supplementary covariate for leader-type heterogeneity tests (Online Appendix C.5).

China Land Market Network (2003–2022). Transaction-level land-sale data; used to verify total-investment null (Proposition 1 null prediction) through an independent margin.

Data provenance, key-field definitions, and linkage rates are in Online Appendix A.2. The full replication archive (code + data + construct-validity outputs) is pre-registered at OSF (DOI 10.17605/OSF.IO/ZMJY5) and mirrored on Zenodo with a reviewer-accessible token.

5. Substantive Applications

Section 4 established that the Visibility Attention Index is a valid measure of compositional policy attention within the domain of Chinese municipal governance rhetoric. This section demonstrates three substantive

applications and reports two pre-registered null results. The purpose of this section is not to make strong causal claims — the instrument is validated, but the surrounding identification architecture in this paper is limited — but rather to illustrate what the instrument can substantively reveal and what it cannot.

5.1 Application I: Compositional Substitution in Accounting Data (P1)

Hypothesis (pre-registered H1): Within-city variation in VAI positively predicts the accounting-based Cosmetic Investment Ratio (CIR) while leaving total infrastructure investment unchanged.

Specification:

$$CIR_{it} = \beta \cdot VAI_{it} + \gamma' X_{it} + \mu_i + \tau_t + \epsilon_{it}$$

where $X_{it} = \{\ln GDP_{pc_{it}}, \ln pop_{it}, ind2-share_{it}\}$, μ_i are city fixed effects, τ_t are year fixed effects, standard errors are clustered by city. Sample: 2,751 city-year observations across 261 cities, 2005–2015 (restricted by MOHURD CIR coverage).

Results (Table 2):

Table 2. P1 — Compositional substitution (CIR ~ VAI, primary specification)

Specification	$\beta(VAI)$	SE	p	95% CI	N
(1) Primary VAI	+0.113	0.037	0.002	[+0.042, +0.185]	2,751
(2) Reconstructed V_ORIG (robustness)	+0.111	0.036	0.002	[+0.041, +0.182]	2,751
(3) Expanded independent VAI_ext	+0.111	0.043	0.011	[+0.026, +0.196]	2,751
(4) Primary VAI, no city FE	+0.089	0.042	0.035	[+0.006, +0.172]	2,751
(5) Primary VAI, no year FE	+0.184	0.055	0.001	[+0.076, +0.292]	2,751

Specification	$\beta(\text{VAI})$	SE	p	95% CI	N
(6) Primary VAI, IV = VAI_lag1	+0.142	0.051	0.005	[+0.042, +0.242]	2,489

All specifications yield $\beta > 0$ at conventional significance and are quantitatively consistent with the pre-registered prediction. The IV specification with VAI_lag1 (a purely mechanical instrument to test attenuation from measurement error) gives $\beta = +0.142$.

Null outcome — total infrastructure investment. Under P1, compositional substitution should not be accompanied by expansion of total spending. We regress $\ln(\text{total urban construction investment})$ on VAI with the same controls and FE structure. The coefficient is $\beta = -0.008$, $p = 0.71$, consistent with the “composition, not expansion” interpretation.

Horse races against alternative explanations. Three standard alternatives do not materially change the estimate (Online Appendix C.2): – GDP–growth incentives: controlling for annual GDP growth gives $\beta(\text{VAI}) = +0.109$, $p = 0.003$. – Urbanization stage: controlling for urbanization rate and its square leaves $\beta(\text{VAI})$ unchanged. – Leader tenure: partialling out party–secretary and mayor tenure years leaves $\beta(\text{VAI})$ unchanged.

Specification–curve analysis. We implement a pre-registered specification curve with 24 permutations (different lexicon thresholds, control–set subsets, panel restrictions). The median β is +0.107, with 24/24 coefficients positive and 22/24 significant at the 5% level. Full curve in Online Appendix B.1.

5.2 Application II: The Within–Document Review–vs–Plan Differential

This section presents what we consider the paper’s novel methodological demonstration. The test exploits the temporal structure of Chinese municipal government work reports — each document contains a *retrospective* past–year review section and a *prospective* next–year plan section — and derives a within–document behavioral prediction from the cadre–attention model.

Theoretical prediction. Under the observability–asymmetry assumption and the cadre–attention model of §2, officials are expected to *emphasize visible achievements* in the retrospective section (where they have discretion over what to highlight from among completed work) more than they *commit to*

visible targets in the prospective section (where functional outputs are specifiable in concrete quantifiable terms). Consequently, the model predicts:

$$VAI_{\text{review}} > VAI_{\text{plan}}$$

at every GWR–level observation, with a magnitude proportional to the legibility asymmetry ϕ and the career–concern weight $(1 - \lambda_i)/\lambda_i$.

Operational implementation. Each GWR is partitioned at the first marker from the set {主要工作安排, 工作思路, 下一步工作, 目标任务, 明年工作} occurring after character 2000. We compute VAI separately on each section. The sample of successful section splits is $N = 4,330$ GWRs.

Results (Table 3):

Table 3. Application II — Within–document retrospective–vs–prospective VAI differential

Metric	Value
Mean VAI(review)	0.615
SD VAI(review)	0.191
Mean VAI(plan)	0.591
SD VAI(plan)	0.145
Mean Δ (review – plan)	+0.0245
Paired t–statistic	8.40
Two–tailed p	6.2×10^{-17}
95% CI for Δ	[+0.0188, +0.0302]

Interpretation. The differential is positive, highly statistically significant, and of a plausible economic magnitude (~20% of a cross–city VAI standard deviation). Critically, the test *cannot* be produced by: – **Lexicon–selection bias**: both sections use the same lexicon. – **Author–style effects**: the same author (the municipal government) writes both sections of the same report. – **Report–length effects**: VAI is defined as a ratio, insensitive to report length. – **City–level unobservables**: the within–document comparison partials these out mechanically.

This is the paper’s primary methodological demonstration. Within–document behavioral signatures of this kind have not, to our knowledge, been previously reported in the text–measurement literature. The instrument construction and the behavioral test are each pre–registered at OSF DOI 10.17605/OSF.IO/ZMJY5.

5.2.1 Robustness

- **Alternative split markers:** repeating the analysis with only the single marker “下一步” and only “主要工作安排” as candidates gives $\Delta = +0.027$ ($t = 9.1$) and $\Delta = +0.022$ ($t = 7.8$) respectively. The result is insensitive to which marker triggers the split.
- **Restricting to long-document sample** (GWRs with $\geq 20,000$ characters, $N = 2,908$): $\Delta = +0.029$, $t = 8.4$.
- **Restricting to single-leader-tenure window** (3-year consecutive observations): $\Delta = +0.022$, $t = 6.1$.

5.2.2 What this does and does not tell us

The differential is compatible with the upward-signaling mechanism in §2, but it is not a unique diagnostic for that mechanism. Alternative explanations that *also* predict the differential include (a) downstream-facing rhetorical strategy toward citizens, (b) lexical availability bias (visible terms are more naturalistically common in retrospective description), and (c) selection effects on what gets completed versus planned. We regard the differential as a robust *measurement-level* fact about how GWRs are written; its *mechanism* is consistent with — but not uniquely identified by — the cadre-attention model.

5.3 Application III: Structural Welfare Calibration

Setup. The cadre-attention model of §2 yields a closed-form welfare-loss expression:

$$W_i = \frac{1}{2} \cdot \frac{(a_i^* - a_i^{SO})^2}{a_i^{SO}(1 - a_i^{SO})}$$

We calibrate using three moments: (i) observed compositional share $a_i^* = 0.538$ (MOHURD yearbook-derived, preferred over text-based because it maps to monetary units); (ii) socially-optimal benchmark $a_i^{SO} \in [0.40, 0.50]$ anchored in MOHURD urban-design standards; (iii) total urban-renewal expenditure $\approx$ ¥3.5 trillion/year (2020–2024 mean, 8% of which is renewal-attributable).

Central result:

$$W = \frac{1}{2} \cdot \frac{(0.538 - 0.450)^2}{0.45 \times 0.55} \approx 0.0156.$$

Applied: $W \times 0.08 \times 3500 \text{ billion} \approx \text{¥4.4 billion per year}$ of compositionally–misallocated investment.

Sensitivity (Table 4):

Table 4. Welfare–loss sensitivity

a^{SO}	Utility	W (¥ billion/yr)
0.50	Log	0.4
0.45	Log	4.4
0.40	Log	11.0
0.45	CRRA($\gamma=2$)	8.8
—	Back–of–envelope shadow–priced	55.0

We report the defensible range as **¥1–¥15 billion per year** under log utility; the BOE upper–bound of ¥55B is a genuine possibility under less–conservative shadow pricing.

5.4 Transparent Reporting of Pre–Registered Null Results

We pre–registered two additional tests at OSF (DOI 10.17605/OSF.IO/ZMJY5). Both return null results. We report them transparently.

5.4.1 Null 1: Event study around central anti–corruption inspections (pre–registered H2)

Pre–registered prediction: central–inspection arrival reduces VAI at event–time $k \geq 0$ with $|\Delta a| \geq 0.02$, under a Callaway and Sant’Anna [13] staggered DiD specification.

Deviation D–B–1: the pre–registered Callaway–Sant’Anna estimator was infeasible given the 2013–2014 rollout structure (two distinct annual cohorts, no never–treated controls after 2015). A TWFE event–study substitute was implemented. **Deviation D–B–2:** a Sun and Abraham [14] heterogeneity–robust alternative was added post–hoc. **Deviation D–B–3:** the analysis window was extended to 2017 using CCDI Rounds 6–9 (including 回头看 re–inspections).

Results (summary):

Estimator	Sample	$\beta(k=0)$	SE	p
TWFE event study	2013–2014 first-treatment only, 2009–2018 window	–0.065	0.025	0.011
Sun–Abraham 5 cohorts	Same	–0.019	0.117	0.87
Sun–Abraham 9 cohorts	2013–2017, includes re-inspections	+0.016	0.117	0.19

The TWFE point estimate under the narrow window gave the pre-registered-direction negative effect, but this does not survive two robustness checks: (i) heterogeneity-robust estimation under the same window attenuates the effect to statistical insignificance, and (ii) extension of the sample to Rounds 6–9 reverses the sign. All nine cohort-specific CATT($k=0$) values under the expanded sample are positive (+0.011 to +0.083).

We interpret H2 as definitively null. The narrow-window TWFE result is a classic staggered-treatment-heterogeneity artifact [15, 16] rather than a causal inspection effect. The null is reported in full detail to conform to pre-registered null-reporting standards.

5.4.2 Null 2: Individual-level CFPS micro-foundation (pre-registered H5)

Pre-registered prediction: CFPS amenity-category-specific satisfaction items show a dual-direction differential — higher satisfaction with visible amenities (parks, roads), lower with functional amenities (water, heating) — at city-year-level VAI, with |Cohen’s d | ≥ 0.10 .

Deviation D–F–1: CFPS amenity-specific items are not released in the public cleaned panel. We substituted three closest outcomes: government evaluation (qn1101), life satisfaction (qn12012), and self-rated health.

Results:

Outcome	$\beta(\text{VAI}_z)$	SE	p	Cohen's d	N
County govt evaluation	−0.010	0.011	0.35	−0.011	62,139
Life satisfaction	−0.002	0.006	0.71	−0.002	64,458
Self-rated health	−0.003	0.008	0.71	−0.002	65,049

All three coefficients are indistinguishable from zero. Two one-sided tests (TOST) at Cohen's $d = \pm 0.10$ equivalence bounds reject $|\beta| \geq 0.10$ for all three outcomes at $p < 10^{-6}$.

What the null does and does not tell us. Within the substituted outcome set, we can reject citizen–popularity effects above $d = 0.10$. This does NOT confirm the upward–signaling mechanism (the original pre–registration specified a positive direction, which failed); it only bounds the magnitude of citizen–level effects conditional on the substitute outcomes. A true test of the pre–registered H5 requires amenity–category–specific items that are not publicly available. We report the null as *exploratory* rather than *confirmatory*.

5.5 Summary

Two positive applications (P1 compositional substitution with $\beta = +0.111$ in accounting data; the novel E–B within–document differential with $\Delta = +0.025$ and paired $t = 8.4$) and one theory–calibrated welfare quantification (¥1–15B/yr) demonstrate that the VAI instrument substantively captures compositional policy attention within the domain of Chinese municipal governance rhetoric. Two pre–registered null results (the inspection event study H2 and the CFPS micro–foundation H5) refine the scope and limit the strength of causal claims the instrument can support. The methodological contribution is the instrument itself, its five–test construct–validity battery, and the pre–registration–and–transparent–null–reporting protocol.

6. Discussion

6.1 What the Paper Establishes Methodologically

This paper's primary contribution is the Visibility Attention Index (VAI) — a validated, pre–registered, replicable text–measurement instrument for compositional policy attention in bureaucratic documents. The instrument survives four of five pre–registered construct–validity tests:

- **Independent–dictionary replication** (E–A, $r = 0.93$ under a 150–term alternative lexicon);
- **Within–document behavioral signature** (E–B, $\Delta = +0.025$, paired $t = 8.4$, $p < 10^{-16}$) — the paper’s novel methodological demonstration;
- **Cross–source accounting correlation** (E–D, $r = 0.24$ with MOHURD yearbook–derived CIR, $p < 10^{-30}$);
- **Specification–curve stability** (24/24 positive coefficients in the primary application, 22/24 significant).

The fifth test, using Chinese Wikipedia as a third–party encyclopedic corpus, returns a null ($r = -0.15$), which we interpret as scoping the instrument to *governance–rhetoric* rather than *encyclopedic–description* domain. This is a bounded finding, not a failure: the instrument is valid within its domain of origin and does not extrapolate to descriptive text about the same referent. We document this scope restriction explicitly rather than claim broader external validity we have not established.

6.2 What Substantively Follows (Conditional on the Instrument)

The applications of VAI demonstrate that:

- Chinese municipal officials systematically allocate text–rhetorical attention toward visible urban–renewal categories ($P1$, $\beta(\text{VAI} \rightarrow \text{CIR}) = +0.111$, $p = 0.002$ under accounting–based CIR).
- This allocation is asymmetric across the temporal structure of the reports themselves — retrospective sections are more visibility–loaded than prospective sections (E–B, within–document paired differential).
- Under a Cobb–Douglas social–welfare model with conservative log–utility, the compositional distortion implies a welfare cost on the order of ¥1–15 billion per year in compositionally–misallocated urban infrastructure investment.

These are substantive findings but all of them are *conditional on the instrument being valid*. The strength of the substantive claims is bounded by the validity–evidence we have presented.

6.3 What We Cannot Establish

Three limitations bound the strength of claims this paper can make.

Limitation 1: Causal identification is not achieved. The pre-registered quasi-experimental test around central anti-corruption inspections does not survive heterogeneity-robust estimation under a sample extended to include the 2016–2017 Rounds 6–9 (including 回头看 re-inspections). The sign reverses from -0.065 in the narrow-window TWFE specification to $+0.016$ in the expanded Sun-Abraham specification, consistent with the narrow-window result being a staggered-treatment-bias artifact. We report this in §5.4.1 as a pre-registered null. Causal identification of what *drives* visibility bias remains an open problem for future work.

Limitation 2: External construct validity is not established. Our five construct-validity tests are internal to the governance-text domain. The pre-registered third-party test using Chinese Wikipedia returns a null, which we interpret as evidence of a domain mismatch between governance rhetoric and encyclopedic description rather than evidence of invalid VAI. A genuinely external test using *policy-rhetoric* third-party text (Xinhua local-policy news; CNKI 重要报纸 political coverage) would require institutional CARSI-authenticated database access and is deferred to future work.

Limitation 3: Micro-foundation is not established. Individual-level CFPS data cannot detect citizen-level satisfaction effects above Cohen's $d = 0.011$. This is a genuine upper bound on citizen-popularity channel effects within the substitute-outcome set we could construct. It does NOT confirm the upward-signaling theoretical mechanism; it simply bounds an alternative. A true test of amenity-category-specific satisfaction effects requires primary survey data collection not available to us.

6.4 Portability to Non-Chinese Settings

The VAI construction protocol is in principle portable to other institutional settings where bureaucratic agents produce structured annual reports:

- **European Union country-level National Reform Programmes:** 27 EU member states submit annual NRPs to the European Commission. A visibility-vs-functionality lexicon adapted to EU-policy vocabulary (e.g., “digitalisation flagship projects” versus “pension-system structural reform”) could be tested.
- **Indian state-level budget speeches:** 28 states, annual, multi-decade, institutionally significant as political signaling.
- **Mexican municipal *Plan de Desarrollo* documents:** Mexico's 2,471 municipalities produce 3-year strategic plans; visibility bias should be detectable in the Mexican municipal-governance context.

In each case, the construction protocol is: (i) seed lexicon from 50 random documents; (ii) inter-coder agreement refinement; (iii) within-document validity test (the review-vs-plan differential would apply to any document with temporal section structure); (iv) cross-source validation against accounting or audit data if available. The core methodological innovation — the behavioral-signature validation (E-B) — does not require translation or domain expertise in the target language and is especially portable.

What is not portable: the cadre-attention model's specific observability-asymmetry parameters (ω_V, ω_F) are institutionally grounded in the Chinese cadre evaluation system and would need re-calibration in each setting. The welfare-calibration constants (α^{SO}, B) are likewise context-specific.

6.5 Policy Implications (Conditional on the Cadre-Attention Model)

If the cadre-attention model is substantively correct — a claim we argue is theoretically motivated and partially supported by our within-document behavioral signature (E-B) but not cleanly identified causally — three policy levers follow.

Lever 1: Increase observability of functional categories (reduce the legibility asymmetry ϕ). Real-time pipe-pressure telemetry, open-access waterworks dashboards, mandatory structural-safety inspection reporting all reduce the ω_V/ω_F ratio. Implementation cost (central MOHURD estimate): ¥10–15B over 10 years.

Lever 2: Introduce functional output targets in cadre evaluation (increase λ). Amending the evaluation formula to include quantitative functional-investment targets reduces the career-concern pressure toward visibility. Administrative cost: modest.

Lever 3: Citizen-audit portals (supplementary pressure independent of career value). Public web portals allowing citizens to score local infrastructure quality. Given our C1 null, this is primarily *informational* rather than directly *incentive-based*.

Under structural welfare-loss calibration, the 3-lever package has a combined net present value of approximately **¥50–60 billion over 10 years**. This is conditional on the cadre-attention model being correct and on the structural parameters we calibrate; we offer it as a policy-implication illustration, not as a policy forecast.

7. Conclusion

We have introduced and validated a text–measurement instrument for compositional policy attention in bureaucratic documents. The Visibility Attention Index, constructed from 6,294 Chinese municipal government work reports, survives four of five pre–registered construct–validity tests, with the fifth test providing a useful scope–bounding null. We demonstrate three substantive applications: compositional substitution in accounting data (P1), the within–document retrospective–vs–prospective behavioral signature (E–B — novel to our knowledge), and structural welfare calibration. We report two pre–registered nulls transparently, including one that reverses a claim supported in the narrow–window TWFE specification but not in the heterogeneity–robust extended specification.

The paper’s primary contribution is methodological: an interpretable, replicable, pre–registered text–measurement tool for compositional policy attention, with explicit domain–scope documentation. The secondary contribution is substantive: first empirical estimates of compositional misallocation in Chinese urban infrastructure under a formal cadre–attention framework.

We invite extension in three directions: (i) construction of a true policy–rhetoric third–party corpus (Xinhua; CNKI 重要报纸) to strengthen external validity; (ii) expansion of the identification strategy to additional CCDI rounds or alternative quasi–experimental shocks; (iii) application to non–Chinese bureaucratic settings (EU National Reform Programmes; Indian state budgets; Mexican municipal plans).

Pre–registration: OSF DOI 10.17605/OSF.IO/ZMJY5. Replication archive: Zenodo DOI 10.5281/zenodo.19569979; GitHub: andyhsi2023–cq/visibility–bias–v2.

Statements and Declarations

Competing Interests. The author declares no competing financial or non–financial interests related to this work.

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Authors' Contributions. Hongyang Xi is the sole author. He conceived the study, constructed the lexicon and instrument, performed all data collection and analysis, drafted the manuscript, and prepared the replication archive.

Data Availability. All data, lexicons, code, and intermediate outputs supporting the findings are archived at Zenodo (DOI [10.5281/zenodo.19569979](https://doi.org/10.5281/zenodo.19569979)) and mirrored on GitHub (andyhsi2023-cq/visibility-bias-v2). The Zenodo record is currently Restricted during peer review; reviewer access is provided via the token in the cover letter and will be flipped to Public upon acceptance. Source government work reports are scraped from publicly accessible municipal government websites; CFPS individual data are obtained from the Institute of Social Science Survey at Peking University under their public data-use agreement. MOHURD Urban Construction Statistical Yearbook (2005–2015) is publicly archived.

Code Availability. All analysis scripts (Python, statsmodels, pandas) are archived in the Zenodo replication record.

Pre-registration. This study is pre-registered at the Open Science Framework (DOI [10.17605/OSF.IO/ZMJY5](https://doi.org/10.17605/OSF.IO/ZMJY5), archived 2026-04-14). All deviations from the pre-registration are logged in Appendix D.

Ethics Approval. Not applicable. The study analyses publicly available bureaucratic documents and de-identified secondary survey data (CFPS); no primary human-subjects research was conducted.

Consent for Publication. Not applicable.

Use of Generative AI. The author used a large language model for AI-assisted copy-editing of draft text (improvements to readability, grammar, and consistency). All substantive content, analysis, and interpretation are the author's own. No AI tool was used to generate or analyse data, draft analytical conclusions, or write analysis code beyond line-completion suggestions.

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